Supplement of: A Simple Method for Helium Step Heat Experiments at High Water or Sulfur Fugacity

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S1 Ampoules and Analytical Details

Heating steps and isotopic measurements for all samples were automated using custom LabView routines with the exception of the tank helium-soaked empty tube experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer after the 7th measurement. For all other ampoules, a Watlow temperature controller held the ampoule at the temperature set point for a pre-programmed duration, after which the ampoule was allowed to cool while the evolved gas was prepared for measurement for about 25 minutes. After the first 7 measurements, the empty tube experiment on the Helix-MC-Plus^{10K} was instead set to an elevated temperature manually while the evolved gas was allowed to accumulate continuously in an attempt to mitigate uncertainties related to frequently changing temperatures. The accumulated gas was measured every 40-120 minutes. For this experiment, the temperature was changed either during the 1 minute pump-out of the lamp cell after inletting gas into the mass spectrometer, or during a period of helium accumulation for measurement. When the temperature was changed during an accumulation period, no permeability was calculated for that step, and the measured abundances of those steps were used only to track the helium contents of the tube.

Helium isotope measurements of evolved gases from step heats of the ampoules equilibrated with 2-20 torr of high purity tank helium ($R \approx 0.15\ R_a$) were acquired at Caltech on a Thermo Fisher Helix-MC-Plus^{10K} mass spectrometer in peak hopping mode. Helium was purified by exposing evolved gases to a charcoal trap held in liquid nitrogen, a titanium sponge getter, then a SAES NP10 getter before freely expanding into the Helix-MC-Plus^{10K}.

Helium-4 evolved during the step heat of the silica glass-encapsulated Durango apatite was quantified using isotope dilution with a high purity 3He spike on a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer. Evolved gas from the ampoule in the lamp heating cell was allowed to mix with the 3He spike and the mixture was frozen on charcoal held at 14 K within a cryostat, then helium was selectively released to the mass spectrometer by raising the temperature to 36 K.

Standard gasses were routinely measured to monitor drift in the sensitivities of 3He and 4He. Blanks are approximately 5 pccSTP ^{4}He, which is very small compared to all interpreted ^{4}He measurements.

Table S1. Silica Glass tubes that were the subject of helium step heat experiments in this work. The ampoule labeled "Quad" had evolved 4He quantified by isotope dilution using a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer, whereas ampoules labeled "MC" had both evolved 3He and 4He quantified using a Helix-MC-Plus^{10K} multi-collector mass spectrometer. Where 3He was measured, the ampoule was soaked in tank helium with an isotope ratio of ~0.15 Ra.

Ampoule Contents	Mass Spec.	Inner Diameter μm	Outer Diameter μm	Geometric factor *G* cm^{-2}	Volume Ratio $\left(\frac{V_{glass}}{V_{gas}}\right)$	Content Mass	Helium Source
Empty	MC	747	1159	3264	1.41	—	Tank He
Sulfur	MC	529–550	835–872	5992	1.5	~0.2 mg	Tank He
Water	MC	487–515	734–776	7789	1.27	~0.5 mg	Tank He
Durango Apatite	Quad	850	1350	2394	1.52	3.18 mg	Radiogenic

S2 Relating the Rate Constant to Glass Permeability and Ampoule Geometry

Following Rogers et al. (1954) and Altemose (1961), steady-state permeation of helium through a glass membrane of area *A* and thickness *l* follows from Fick's law with a Henrian solubility *S* (Shelby, 1976). With the permeability $K = DS$, the atom flux *F* through the membrane is:

$$F = -\frac{KA\Delta P}{l} \tag{S1}$$

Recalling that F is an atomic flux in units of $\frac{atoms}{s}$ that could be notated as $\frac{dn}{dt}$:

$$\frac{dn}{dt} = -\frac{KA\Delta P}{l} \tag{S2}$$

We assume that the helium partial pressure in the ultra-high vacuum (UHV) heating cell surrounding the ampoule is 0, so that the pressure gradient equals the partial pressure within the ampoule ($\Delta P = P$). This is a valid assumption for this work because the heating cell is 4-5 orders of magnitude more voluminous than the glass ampoule.

Helium is well-approximated as an ideal gas at experimental conditions, so the ideal gas law (PV = nRT) relates the pressure, *P*, to the atomic abundance, *n*, yielding Eq. (S3):

$$\frac{dn}{dt} = -\frac{AKnRT}{lV} \tag{S3}$$

where *R* is the gas constant $\left(1.36 \times 10^{-22} \frac{cm^3 atm}{K\ atoms}\right)$, *T* is the temperature in Kelvin, and *V* is the internal volume of the ampoule. Defining a geometric factor $G = A/(lV)$ in units of cm^{-2} gives

$$\frac{dn}{dt} = -GKnRT \tag{S4}$$

Where *n* is the number of helium atoms, *t* is the time, *G* is the geometric parameter in units of cm^{-2}, which is a function only of the geometry of the glass membrane, *K* is the permeation constant for helium in fused silica, which follows the Arrhenius law: $\left(K(T) = K_0 e^{-\frac{E_a}{RT}}\right)$ and has been repeatedly measured for fused silica (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968), *R* is the gas constant, and *T* is the temperature in Kelvin. Equation (S4) is identical to the permeation equation used in early studies (Norton, 1953; Altemose, 1961; Swets et al., 1961; Studt, 1962; Masaryk, 1968), except that it is formulated in terms of absolute atomic abundances instead of helium gas pressure by the ideal gas law. It is easily integrated by separation of variables to yield $n_t = n_0 e^{-GKRTt}$. Comparison with main-text Eq. (2) gives $\lambda = KGRT$.

For non-planar membranes, the effective area of permeation, *A*, is not straightforward to evaluate because the surface area on each side of the membrane differs. Masaryk (1968) gives the effective area of permeation for a cylinder (such as for this experiment) as:

$$A = \frac{2\pi hl}{\ln\left(\frac{b}{a}\right)} \tag{S5}$$

Where *a* is the cylinder's internal radius, *b* is the cylinder's external radius, and *h* is its length. This effective area of permeation (Eq. (S5)) can be used with the equation for the internal volume of the cylinder: $V = h\pi a^2$ to evaluate the geometric factor *G* for a cylinder in terms of the ampoule geometry. Reducing yields:

$$G = \frac{2}{a^2 \ln\left(\frac{b}{a}\right)} \tag{S6}$$

Interestingly, *h* cancels so that the permeation rate has no dependence on the ampoule length when permeation through the top and bottom is negligible. Eq. (S6) displays the intuitive behavior that thinner walls and narrower tubes lead to more efficient permeation.

S3 A Numerical Model for Diffusive Permeation of Helium Through Silica Glass Ampoules

Most of the permeation measurements performed on the Helix-MC-Plus^{10K} mass spectrometer were complicated by additional permeation occurring at variable temperature during long cooling times following high temperature step. Such a complication can be effectively avoided by limiting the duration of time for which the diffusion cell is opened to the rest of the vacuum line, or by keeping the sample at-temperature during this equilibration time; however, it is addressed here numerically. We use a numerical diffusive permeation model is used to predict the ampoule's expected permeation fraction (n_t/n_0) for each step with a complex thermal history. This method is also used to

To simulate permeation during cooling of an ampoule, the temperature is decreased at every step following a parabolic rate law determined using the Watlow temperature records of the cooling ampoule. These functions and the thermal data to which they are fit are reported in the dataset (https://doi.org/10.5281/zenodo.23872881). Further details are given by Thyer (2026).

We numerically solve the radial diffusion equation for cylinders using the Forward-Time Central-Space (FTCS) Finite Difference Method (FDM) along a radial transect of a model ampoule of any geometry for small, discrete time intervals, neglecting permeation through the ends. The system is spatially discretized into 100 cells along a radial transect, and numerical stability was ensured by solving for the time interval of a single step, Δt, for $\frac{D\Delta t}{\Delta x^2} = 0.5$, to satisfy the Courant-Friedrichs-Lewy (CFL) condition for numerical stability.

Helium concentrations along the transect (except for the boundaries, where r = a and r = b) are updated according to the following equation, an FTCS discretization of the radially symmetric diffusion equation for a cylinder:

$$\frac{dC_r}{dt} = D\left[\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r}\right]$$

$$\Delta C_r \approx \Delta t \bullet D\left[\frac{C_{r-\Delta r} - 2C_r + C_{r+\Delta r}}{\Delta r^2} + \frac{1}{r}\frac{C_{r+\Delta r} - C_{r-\Delta r}}{2\Delta r}\right] \tag{S7}$$

Dirichlet boundary conditions were enforced at the inner and outer boundaries. The concentration at the model ampoule's outer boundary (where r = b) was held at 0, while the concentration at the inner boundary (where r = a) was calculated at each discrete step by multiplying the solubility (in units of $\frac{atoms}{atm \times cm^3}$) by the helium partial pressure within the ampoule, according to Eq. (S8).

$$C_{He}^{glass} = S(T) \bullet P_{He}^{gas} \tag{S8}$$

In diffusive steady state, these boundary conditions give the concentration profile across the ampoule wall (Thyer, 2026):

$$C(r) = S(T) \bullet P_{He}^{gas} \bullet \frac{\ln\left(\frac{r}{b}\right)}{\ln\left(\frac{a}{b}\right)} \tag{S9}$$

It is sometimes useful to normalize the helium concentration of the glass to the concentration of helium in the gas. To do this, the internal helium partial pressure is related to an atomic or molar concentration using the ideal gas law:

$$C_{He}^{gas} = \frac{n}{V} = \frac{P_{He}^{gas}}{RT}$$

The pressure term vanishes in the ratio of the two concentrations and depends only on temperature:

$$C_{He}^{glass/gas} = \frac{C_{He}^{glass}}{C_{He}^{gas}} = S(T) \bullet RT \tag{S10}$$

Eq. (S10) strongly favors the gas and is only a weak function of temperature, varying between 2.5% and 2.8% between 0 and 300°C.

Fluxes between the glass and the gas inside and outside the ampoule are evaluated with finite differences at the inner and outer walls and used to update the helium contents of the ampoule and of the infinitely diffuse measureable resevoir outside the ampoule at each time step (Thyer, 2026).

The simulation of encapsulated titanite in main text Fig. 4 uses a much simpler model, because the temperature does not change with time: therefore the helium concentration profile need not be tracked, and is assumed to be in equilibrium with the helium inventory of the gas phase. For each timestep, the helium content of the ampoule is updated according to the differential equation $dn/dt = \Omega - \lambda n$. where Ω is the helium production rate of the encapsulated titanite, which is evaluated following the forward equations of **?**, and λ is the permeation rate constant for the ampoule at the given temperature, which is evaluated as $\lambda = KGRT$ where *G* is the geometric factor of a model ampoule with inner diameter of 0.8 mm and outer diameter of 1.0 mm and *K* is the permeability of helium in fused silica at 300°C from Swets et al. (1961).

S4 Derivation of the Bounding Method

For long heating steps ($\lambda t \gtrsim 5$), the helium evolution of the sample can be approximated as the measured value ($\omega \approx m$). The uncertainty of this approximation due to permeation lag can be bounded from above by calculating the maximum possible helium residing in the tube between steps, i.e. $n_0 \leq n_0^{UB}$ and $n_0 \leq n_0^{UB}$ and $n_t \leq n_t^{UB}$. By Eq. (1) (main text), these bound the encapsulated material contribution ω within:

$$m - n_0^{UB} \leq \omega \leq m + n_t^{UB}. \tag{S11}$$

A new first order differential equation like Eq. (S4) is formulated to describe the evolving helium inventory of the ampoule when the encapsulated phase is evolving helium. We assume a constant source term (Ω) to conservatively approximate (overestimate) the helium influx into the ampoule from the degassing natural sample.

$$\frac{dn}{dt} = \Omega - \lambda n \tag{S12}$$

Where $\Omega = \frac{\omega}{t}$ and $\lambda = KGRT$. Immediately, the steady state quantity of helium atoms (n_{ss}) present in the ampoule can be found by solving Eq. (S12) for the steady state condition where $\frac{dn}{dt} = 0$.

$$n_{ss} = \frac{\Omega}{\lambda} = \frac{\omega}{\lambda t} \tag{S13}$$

Additionally, the helium concentration within the ampoule during the approach to the steady state concentration can be solved for by integrating Eq. (S12) with respect to *t*. First, the integrating factor for Eq. (S12) is identified: $I = e^{\lambda t}$, which is multiplied to both sides to yield Eq. (S14).

$$e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n = e^{\lambda t}\Omega \tag{S14}$$

Then both sides are integrated with respect to *t:*

$$\int \left(e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n\right) dt = \int e^{\lambda t}\Omega dt$$

Because $\frac{d}{dt}(ne^{\lambda t}) = e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n$ by the product rule Eq. (S14) reduces to:

$$n_t e^{\lambda t} = \frac{\Omega}{\lambda}e^{\lambda t} + C \tag{S15}$$

Where *C* is an unknown constant introduced by indefinite integration. This expression becomes more intuitive when both sides are divided by the integrating factor and by substituting the steady state concentration from Eq. (S13).

$$n_t = n_{ss} + Ce^{-\lambda t} \tag{S16}$$

The integration constant *C* can be solved by enforcing the condition that $n_t(t=0) = n_0$, leading to:

$$n_0 = n_{ss} + C \tag{S17}$$

Solving Eq. (S17) for *C* and substituting into Eq. (S16) yields Eq. (S18). Now an inequality is assigned to the equation to reflect the initial assumption that the isothermal helium outgassing rate (Ω) is constant *or decreasing*.

$$n_t \leq\ n_{ss} + (n_0 - n_{ss})\, e^{-\lambda t} \tag{S18}$$

For long timescales (i.e. when $e^{-\lambda t} \approx 0$), Eq. (S18) reduces to $n_t \leq\ n_{ss}$. This is effectively true for steps long enough such that $t \geq \frac{5}{\lambda}$, for which $e^{-\lambda t} \leq 0.007$. It is advisable to only consider the bounding method for steps of sufficiently long duration such that $t \geq \frac{5}{\lambda}$ so that the estimation for n_t can be reduced to simply the steady state value and not be dependent upon the estimation for n_0 and its uncertainties.

To estimate the upper bound on the permeation lag (n_t^{UB}), one must consider the circumstances that would produce the highest possible value for n_t. Thus, we assume the upper bound estimate for ω, which is given in Eq. (S11) as $m + n_t^{UB}$. Substituting $\omega^{UB} = m + n_t^{UB}$ into Eq. (S13) to express n_t^{UB} in terms of the measured value *m* for long steps with $\lambda t \geq 5$:

$$n_t^{UB} = n_{ss}(\omega^{UB} = m + n_t^{UB}) = \frac{m + n_t^{UB}}{\lambda t}$$

Solving for n_t^{UB} yields Eq. (S19):

$$n_t^{UB} = \frac{m}{\lambda t - 1} \tag{S19}$$

S5 Comparison to Literature Permeability Values

Predicted and measured fractional yields for all permeation steps in ampoules besides the encapsulated water experiment agree well (within 1σ: Figs. S1, S2) with the predictive equations presented in this work and with the permeabilities reported by Swets et al. (1961) and Jones (1953) for all studied temperatures (24-300°C) and step fractional yields (up to ~80%). One outlier measurement on the empty tube experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer was excluded from Figs. S1 and S2.

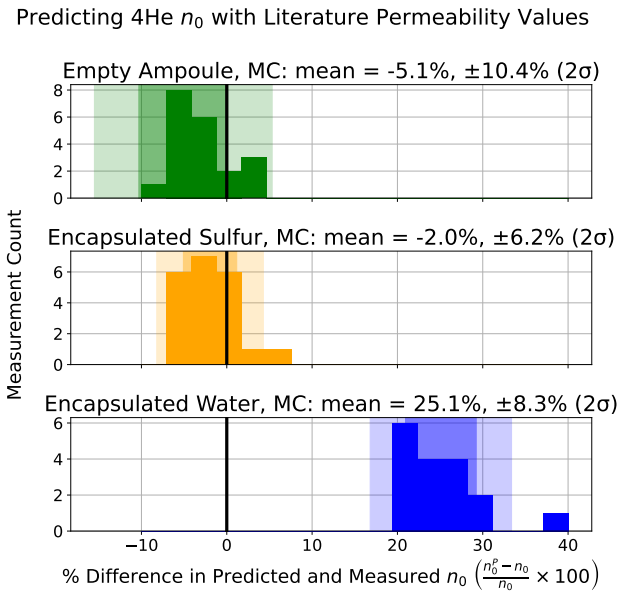


Figure S1. Histograms showing the relative residuals of 4He n_0 predictions using measured geometry and the helium permeabilities of fused silica glass from Swets et al. (1961). These predictions were evaluated analytically from main-text Eq. (3) or numerically with the numerical diffusion model described above. The shaded regions behind the histograms correspond to the 1 and 2 sigma (σ) standard deviations of the distributions. The measured n_0 values of the empty and sulfur-filled ampoules are within 1σ of the predicted values for the empty and sulfur-filled tubes. On the other hand, measured and predicted 4He n_0 for water-filled ampoules are well resolved from a 0 residual beyond 2σ: they are not predicted by the same framework that explains the other ampoules. Internal reproducibilities (1σ) of a gas standard repeatedly analyzed during these experiments were better than 1% for 4He, requiring that most of the scatter in these data be related to factors outside of analytical precision, such as long/inconsistent ramp times during heating or uncertainties or rounding of heating times.

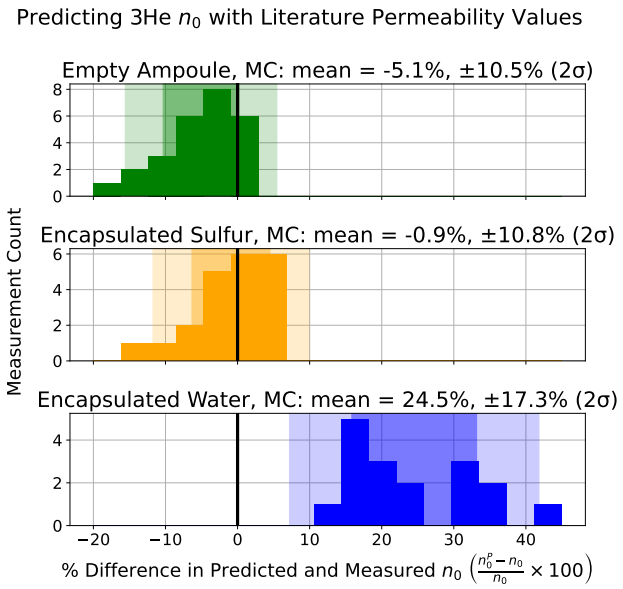


Figure S2. Similar to Fig. S1, but with 3He permeabilities taken from Jones (1953) and Swets et al. (1961).

S6 Additional Analytical Corrections for the Empty Ampoule Calibration Experiments

S6.1 Anomalous Signal Recalculation for the Tank Helium-Soaked Empty Ampoule

The last high temperature step of the empty ampoule experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer had an unusual and non-linear signal evolution at the beginning of the measurement for both 3He and 4He. For this measurement, 4He was recalculated using a linear regression on the last three of six measurement cycles with an R^2 of 0.985 to regress to the 4He content at time 0. Typically, the signal evolution for a measurement like this would have a negative slope because the helium in the mass spectrometer is consumed as it is ionized and measured, but the slope for this measurement was positive. 3He from this measurement was calculated based on the recalculated 4He and an assumed isotope ratio for the measurement of 0.07 Ra. Three more measurements were taken over the next few days to ensure complete degassing of the ampoule. Two of these measurements yielded helium above blank, but were small compared to the final high temperature step. The yields of these steps were all considered in the calculation of the experiment's total yield. This final step does not affect the in-situ fits of the main text, but it does affect the independently known helium inventory of the ampoule backcalculated from the total yield.

S6.2 Independent Verification of the Encapsulated Durango Apatite Mass

The total yield for the encapsulated Durango apatite experiment (main text, §3.3) is estimated from the precisely-measured mass of the encapsulated apatite (3.18 mg) and a helium concentration of 194.2 nccSTP/mg (Farley, 2000). This concentration

Table S2. In-situ permeation rate constants of the empty ampoule, fit to consecutive isothermal permeation steps with Eq. (6). The first steps of the 125 and 200 °C runs are excluded (see main text), leaving exactly determined two-step fits. The last two 250 °C accumulations (steps 28–29) are also excluded. n_0 (true) is known independently from the experiment's total yield but is not used in the fits. λ_{pred} is predicted from the measured ampoule geometry and literature permeabilities (Swets et al., 1961; Jones, 1953) and is shown only as a check. Uncertainties are 1σ from the fit covariance.

Isotope	T (°C)	Steps	λ ($10^{-5}\,\mathrm{s}^{-1}$)	$\lambda/\lambda_{\mathrm{pred}}$	n_0 fit / true	λt_{cum}	$\lambda(^3\mathrm{He})/\lambda(^4\mathrm{He})$
^{4}He	125	2 (8–9)	2.063 ± 0.235	0.945	1.036	0.13	
^{3}He	125	2 (8–9)	2.075 ± 0.472	0.873	1.110	0.13	1.006 ± 0.256
^{4}He	150	3 (11–13)	3.110 ± 0.074	0.930	1.033	0.40	
^{3}He	150	3 (11–13)	3.525 ± 0.139	0.964	1.014	0.45	1.133 ± 0.052
^{4}He	175	4 (14–17)	4.662 ± 0.040	0.951	0.999	0.59	
^{3}He	175	4 (14–17)	5.273 ± 0.108	0.980	0.981	0.67	1.131 ± 0.025
^{4}He	200	2 (20–21)	5.825 ± 0.063	0.841	1.122	0.44	
^{3}He	200	2 (20–21)	6.277 ± 0.392	0.823	1.167	0.48	1.078 ± 0.068
^{4}He	250	6 (22–27)	12.216 ± 0.013	0.969	0.996	2.33	
^{3}He	250	6 (22–27)	14.233 ± 0.137	1.020	0.986	2.72	1.165 ± 0.011

was independently verified by the measurement of 8 different >0.5 mg aliquots of the same Durango apatite sample in-lab undertaken by the first author, which yielded concentrations of 194 ± 9 nccSTP/mg.

S7 In-situ Measurements of the Rate Constant λ of Helium Permeation

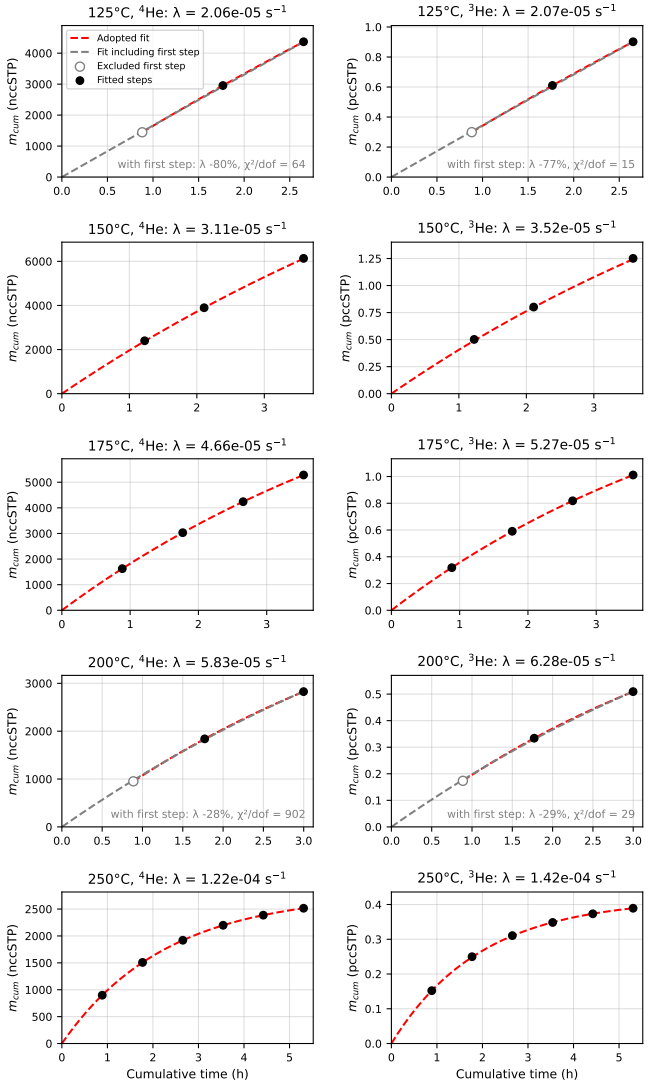


Figure S3. Cumulative yield against cumulative time for each isothermal run of permeation steps of the empty ampoule (rows: temperature; columns: isotope), with the rate constants fit in situ using Eq. (6) (red dashed curves). Hollow grey points are the first steps of the 125 and 200°C runs, which are excluded because the glass had not yet reached a diffusive steady state; the red curves for those runs are fit to the remaining two steps, and the grey dashed curves are the fits that include the first step (annotated with the change in λ and the resulting χ^2/dof).

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